

WASP-43b

R-0300

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_260403 · created 2026-07-17T20:13:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461133.7083 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.033
Transit depth (Rp/Rs)^2	3.6 +/- 1.26 [%]
Orbital Inclination (inc)	79.63 +/- 2.25
Airmass coefficient 1 (a1)	0.85 +/- 0.013
Airmass coefficient 2 (a2)	0.117 +/- 0.086
Scatter in the residuals of the lightcurve fit is	1.56 %
Best Comparison Star	#3 - [363, 290]
Optimal Method	PSF photometry
Transit Duration (day)	0.041 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.19 ± 0.033 vs 0.1594 (deviation 0.93σ)
Duration fit vs published	0.041 d vs 0.0512 d (deviation 0.73σ)
Mid-time O–C	4.8 min
Depth SNR	5.8
Residual scatter	1.56 %

Light curve

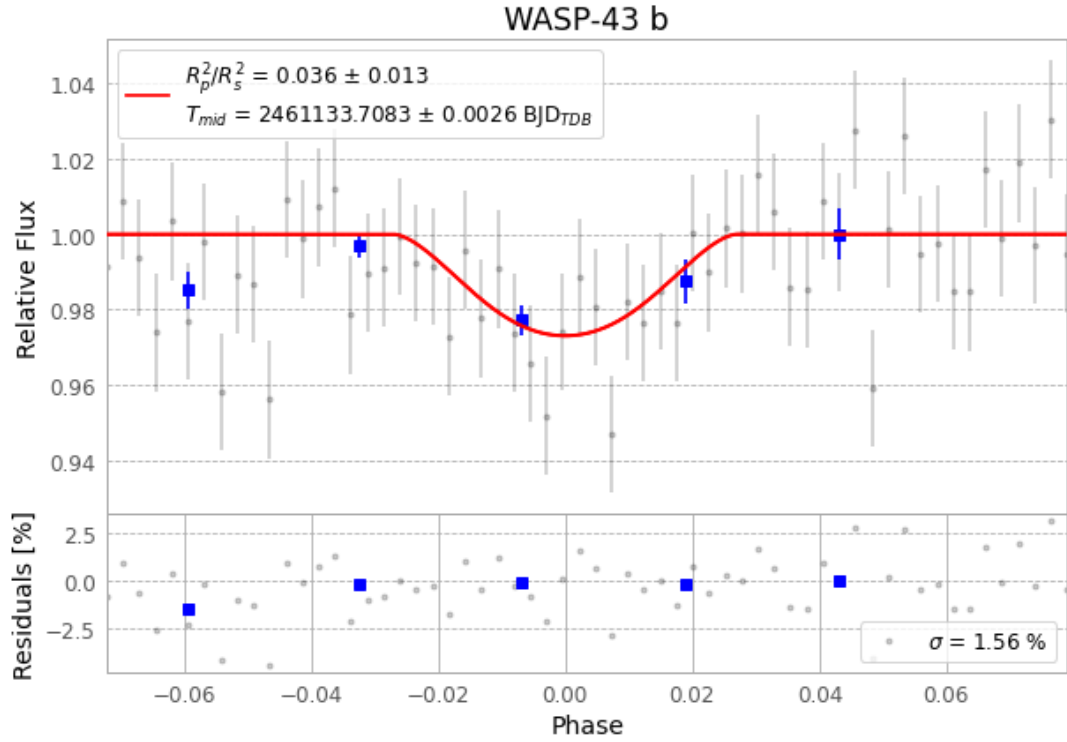


Figure 1 — FinalLightCurve_WASP-43 b_2026-04-02.png (SHA-256 304db135159cd9ee...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 163], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0316fd57727a6144...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 7f15174

Data provenance

60 FITS frames (39 MB), WASP-43260403032716.FITS → WASP-43260403062416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260403.log (b20ddf83aed2...), FinalLightCurve_WASP-43 b_2026-04-02.png (304db135159c...), FinalLightCurve_WASP-43 b_2026-04-02.csv (6fc8d78e4dbb...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 20:13Z.